Ensemble Learning :-

taking knowledge from different people

Ensemble → collection of ~~(lot)~~ learners

Two approaches :-
1. Bagging
2. Boosting

Since, decision tree was overfitting so, people came up with ensemble.

1. Bagging Approach :- [Bootstrapped Aggregation]

x→ design matrix

x are the same samples taken from a large dataset

$$X_1 = \begin{bmatrix} - x^{(1)} - \\ - x^{(2)} - \\ \vdots \\ - x^{(n)} - \end{bmatrix}_{n\times d} \quad y_1 = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y \\ \vdots \\ y^{(n)} \end{bmatrix}_{n\times 1} \Rightarrow f_1(x;\theta)$$

$$X_2 = \begin{bmatrix} - x^{(1)} - \\ - x^{(2)} - \\ \vdots \\ - x^{(n)} - \end{bmatrix}_{n\times d} \quad y_2 = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}_{n\times 1} \Rightarrow f_2(x;\theta)$$

B → no. of datasets we can take from the parent large dataset

B dataset
↳ bag

$b = \{1, 2 \dots B\}$.

$$\underbrace{f_1(x_1;\theta_1)}_{\text{each is a model trained for different datasets}},\ f_2(x_2;\theta_2),\ f_3(x_3;\theta_3) \dots f_B(x_B;\theta_B)$$

all these f are models, different models for different datasets.

aggregate or avg of the output of all models

$\cancel{f(x;\theta) = \frac{1}{B}\sum_{i=1}^{B} f_i(x_i;\theta_i)}$ · Aggregation:

$$\boxed{f(x;\theta) = \frac{1}{B}\sum_{i=1}^{B} f_i(x_i;\theta_i)} \longrightarrow \text{Regression}$$

$$f(x;\theta) = \text{Majority voting}\left(f_i(x_i;\theta_i)\right);\ i \in \{1, B\}$$

classification for regression setting

✱ Limitation of bagging:
- Hard to get so many datasets, one dataset for each model

$$X_1 = \begin{bmatrix} -x^{(1)}- \\ -x^{(2)}- \\ -x^{(3)}- \\ \vdots \\ \vdots \\ -x^{(n)}- \end{bmatrix}_{n\times d} \qquad y_1 = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \\ \vdots \\ \vdots \\ y^{(n)} \end{bmatrix}_{n\times 1}$$

- Bootstrapping : / ~~Boosting~~ Bagging # sample from X_1 with replacement

we randomly put sample in different datasets

create multiple datasets

$$X_2 = \begin{bmatrix} -x^{(3)}- \\ -x^{(n)}- \\ -x^{(3)}- \\ \vdots \\ -x^{(2)}- \end{bmatrix}_{n\times d} \quad y_2 = \begin{bmatrix} y^{(3)} \\ y^{(n)} \\ y^{(3)} \\ \vdots \\ y^{(2)} \end{bmatrix}_{n\times 1} \quad X_3 = \begin{bmatrix} -x^{(2)}- \\ -x^{(2)}- \\ -x^{(1)}- \\ \vdots \\ -x^{(n)}- \end{bmatrix}_{n\times d} \quad y_3 = \begin{bmatrix} y^{(2)} \\ y^{(2)} \\ y^{(1)} \\ \vdots \\ y^{(n)} \end{bmatrix}_{n\times 1}$$

$\vdots$

X_B

$\boxed{X_1}$ # original dataset

$\boxed{X_2}\ \boxed{X_3} \cdots \boxed{X_B}$ ⟩ → These contain random samples from X_2 to X_B

} → X_1 and all bootstrapped datasets will have same no. of samples.